



Study on cutting force of ultrasonic cutting disc cutter for Nomex honeycomb material

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Abstract

To systematically investigate the specific factors influencing cutting force variations in ultrasonic cutting of Nomex honeycomb materials with disc cutters, a comprehensive theoretical framework was developed. This framework incorporates cutting parameters like feed rate, cutting width, and ultrasonic amplitude to establish a theoretical cutting force formula. Through an analysis of ultrasonic vibration-assisted cutting motion trajectories with a disc cutter, a microscopic model of material removal for Nomex honeycomb cutting was developed. This model decouples the isotropic motion of the disc cutter, yielding theoretical cutting force formulas for different directions, elucidating the reduction in cutting force under ultrasonic influence in both the vertical and feed directions. The formulated theoretical cutting force expressions were verified through ultrasonic cutting experiments. Results indicate that when cutting Nomex honeycomb materials with a disc cutter in the vertical direction, cutting force reduction arises from narrowing the effective cutting width between the tool and the material, and from intermittent contact that minimizes tool-material interaction time. A corresponding cutting force expression for this direction was established. In the feed direction, a mechanism similar to intermittent cutting with a straight blade cutter is observed. Ultrasonic vibration's impact stress induces tensile stress in the uncut section, prompting microcrack propagation in the material's front end and reducing fracture stress, thus diminishing the cutting force. A damage factor was introduced to establish the cutting force expression in the feed direction. Error analysis comparing theoretical and experimental cutting force values under varying parameters revealed average errors of 20.75%, 15.5%, and 12.17% for different feed rates, ultrasonic amplitudes, and cutting widths. These findings provide theoretical backing for optimizing cutting parameters.

Keywords Disc cutter · Nomex honeycomb material · Cutting force · Feed rate · Ultrasonic amplitude · Cutting width

1 Introduction

Nomex honeycomb composite materials are an anisotropic composite material made of aramid insulating paper impregnated with phenolic resin and having a hexagonal prismatic lattice axial structure. It has the advantages of high specific strength, high specific stiffness, corrosion resistance, and good insulating properties and is widely used in the fields of aerospace, rail transportation, and shipbuilding [1, 2]. Due to its special anisotropic material characteristics and ortho-hexagonal cell structure, the conventional high-speed milling process of Nomex honeycomb composite materials often results in high dust pollution and poor surface

quality. The ultrasonic machining method has been applied to Nomex honeycomb materials processing, with ultrasonic vibration-assisted machining offering the advantages of reducing cutting forces and improving surface quality [3]. Ultrasonic cutting involves applying high-frequency vibration to the cutting tool based on traditional cutting processes. During the cutting process, the high-frequency vibration of the tool impacts the material, altering the fracture pattern during material removal. Cutting with ultrasonic assistance represents a composite cutting motion that combines micro-cutting with high-speed impact cutting. The high-frequency vibration alters the cutting principles of traditional machining, making the material easier to remove [4].

The cutting tools commonly used for ultrasonic vibration-assisted machining of honeycomb materials are generally divided into two types: straight blade cutters and disc cutters. Straight blade cutters are employed for the removal of large volumes of material during rough machining, while

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disc cutters are mostly used for precision machining of honeycomb surfaces. Due to the simpler motion analysis of straight edge cutters compared to disc cutters, a large number of scholars have conducted research on the cutting mechanism of straight edge cutters.

Based on the theory of fracture mechanics, Huang Xiuxiu [5] introduced the dynamic stress intensity factor to analyze the fracture of materials under ultrasonic cutting conditions using straight blade cutters. It was found that the fracture of materials was caused by the impact of ultrasonic vibration, which led to the premature extension of internal microcracks in the material, enabling the straight blade cutter to remove the material with relatively small force. Based on a simplified model of honeycomb core cutting using straight blade cutters, Wang Yidan et al. [6] established a dynamic model for ultrasonic-assisted cutting based on brittle fracture mechanics theory. It was proposed that the relative magnitudes of the amplitude and the critical elastic deformation of the material determine the form of action between the straight blade knife and the material. Hu et al. [7] and Xiuxiu et al. [8] applied fracture mechanics and tool kinematics theory to consider the shear fracture behavior and separation resistance of materials under the action of cutting tools. They established a cutting force model for ultrasonic cutting with straight blade cutters, which included the forward tilt angle, the lateral tilt angle, the tool wedge angle, the cutting depth, and the feed rate. Xiang [9] established a cutting force model based on the sliding effect, representing the cutting edge wedge angle through a formula that incorporates cutting parameters such as feed speed, tilt angle, and deflection angle. Experimental validation demonstrated that, compared to conventional cutting, the sliding effect induced by ultrasonic vibration can reduce the cutting force. Yuan Xinman [10] analyzed the variation of cutting force under different cutting depths, feed rates, and forward tilt angle, lateral tilt angle by straight blade cutter in ultrasonic cutting experiments of Nomex honeycomb core with straight blade cutters.

Shengyou et al. [11] established a theoretical model of ultrasonic cutting force of disc cutter based on metal cutting theory, indicating that the intermittent cutting effect of the disc cutter on the material is the main reason for the reduction of cutting force, and gave the conditions for achieving intermittent cutting. Meng Qian [12] based on the analysis of the kinematics and cutting force theory of the disc cutter studied the contact and force separation process between the tool and the material within a single cycle after applying ultrasonic vibration, pointed out that the ultrasonic vibration can accelerate the initiation of initial cracks before tool entry, and the impact effect after tool entry can help crack propagation. Sun et al. [13] used finite element simulation to study the ultrasonic cutting process of the disc cutter cutting aluminum honeycomb material. Based on the theory of plastic strain energy, a plastic deformation model of honeycomb

wall was established. The disc cutter alternately cuts the honeycomb wall up and down, causing plastic deformation along the upper and lower surfaces of the tool, thereby forming the cutting and machined surfaces. Jinglu et al. [14] conducted ultrasonic cutting honeycomb core experiments with a disc cutter under no-rotation conditions and used regression analysis to obtain the empirical formula for the cutting force. It was pointed out that the surface quality under ultrasonic cutting is better than that without ultrasonic vibration, and increasing the ultrasonic amplitude can reduce the cutting force. Ahmad et al. [15] conducted a series of single-factor and four-factor level experiments to obtain the conclusion that cutting force increases with the increased feed rate and cutting depth and decreases with the increase of spindle speed and amplitude. Among all the processing parameters, cutting width has a relatively large impact on cutting force. Based on the response surface method, Yuan Xinman [16] established empirical equations for cutting parameters including cutting depth, cutting width, and feed rate. Based on the experimental results, he proposed that cutting width has the greatest impact on cutting force, while cutting depth has the smallest impact on cutting force. Feng et al. [17] proposed a force prediction model for the ultrasonic vibration-assisted milling process, which takes into account three criteria for tool-workpiece separation. Through kinematic analysis, the cutting rotational angles under different motion states were calculated, and combined with the constitutive equations, shear stress was obtained. Subsequently, through coordinate transformation, the feed force, cutting force, and axial force were derived.

Liu [18] established an axial cutting force model that takes into account the workpiece contact and the geometric characteristics of the tool. He proposed a force balance method that considers the acceleration of the chip, the shear plane, and the cutting face, utilizing analytical methods to predict the cutting force.

Currently, the cutting force models for milling cutters and straight-edge tools used in ultrasonic vibration machining are relatively well developed. However, the modeling research for the cutting force of disc cutters is still in its preliminary stages, with many existing models being empirical in nature. Given that the application of disc cutters in ultrasonic vibration machining has become quite common, there is a pressing need to establish a comprehensive model for predicting the cutting force of disc cutters.

This study analyzes the displacement trajectory of the disc cutter under ultrasonic vibration. Based on the motion trajectory, the material removal mechanism of the disc cutter during one vibration cycle is examined. Cutting force models for the ultrasonic vibration disc cutter in the Z direction and X direction are established based on the vibration equations and the material removal mechanisms, respectively. The cutting force models include process parameters such as feed

rate, cutting width, and ultrasonic amplitude. Furthermore, single-factor experiments on feed rate, cutting width, and ultrasonic amplitude were conducted to study their effects on cutting force. The accuracy of the established cutting force formula was verified by experiments.

2 Analysis of motion in disc cutter ultrasonic vibration cutting

2.1 Motion trajectory of disc cutter under ultrasonic vibration

The cutting motion of the disc cutter under ultrasonic vibration is a complex, multi-directional motion that involves both feed motion along the X -axis and ultrasonic vibration motion along the direction of the tool axis and rotational motion along the tool axis.

Figure 1 illustrates the cutting process of Nomex honeycomb material using a disc cutter. During the cutting process of Nomex honeycomb, the chips flow over the surface of the disc cutter, and the upper surface of the cutter is defined as the front cutter face, while the lower surface of the disc cutter is defined as the rear cutter face relative to the already processed surface. To reduce the friction between the rear cutter face and the machined surface of the material, a forward inclination angle β is employed. Additionally, a lateral inclination angle γ is used to ensure that the lowest point of the disc cutter reaches the desired depth.

The motion equation of the ultrasonic-assisted vibrating disc cutter is:

$$\begin{cases} S_X = A \sin(2\pi ft) \sin \beta + V_0 t + R \cos \omega t \\ S_Y = A \sin(2\pi ft) \sin \gamma + R \sin \omega t \\ S_Z = A \sin(2\pi ft) \cos \beta \cos \gamma \end{cases} \quad (1)$$

where:

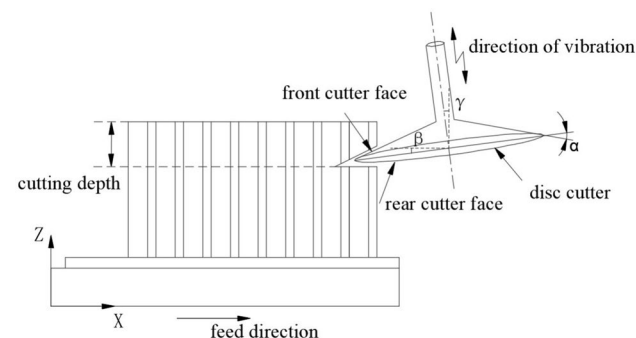


Fig. 1 Schematic diagram of ultrasonic cutting of Nomex honeycomb by disc cutter

S_X, S_Y, S_Z : Displacement trajectory of the disc cutter, mm
 A : Ultrasonic amplitude, mm
 f : Vibration frequency, Hz
 t : Time, S
 ω : Spindle speed, r/min
 V_0 : Feed rate, mm/min
 R : Radius of the disc cutter, mm
 β : Forward tilt angle of the disc, °
 γ : Lateral tilt angle of the disc cutter, °
 α : Wedge angle of disc cutter, °

The rotation of the spindle does not affect the overall tool path of the disc cutter, so the rotation of the spindle is ignored. The other parameters of the motion equation are shown in Table 1.

The geometric structure of the disc cutter used in this experiment is shown in Fig. 2, and the structural parameters are shown in Table 2.

where:

B_1 : Mounting hole diameter, mm
 B_2 : Bump diameter, mm
 R : Radius of the disc cutter, mm
 g : Blade thickness, mm
 G : Total thickness of disc cutter, mm
 α : Wedge angle.

The motion trajectory of the disc cutter is obtained according to the motion equation, as shown in Fig. 3.

2.2 Cutting process in one vibration cycle

Based on the motion trajectory of the disc cutter, the cutting process of Nomex honeycomb material by the disc cutter within one vibration cycle can be obtained, as shown in Fig. 4.

The disc cutter cuts into point A of the honeycomb wall and then moves towards point B. At this point, the rear cutter face separates from the already processed surface, and only the tip of the cutter is in contact with the material. During this process, the tip of the cutter exerts an upward

Table 1 The various parameters of the motion equation

Parameters	f	β	γ	V_0
Value	20,000	1.5	1.5	1000

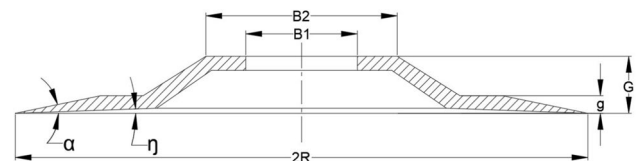
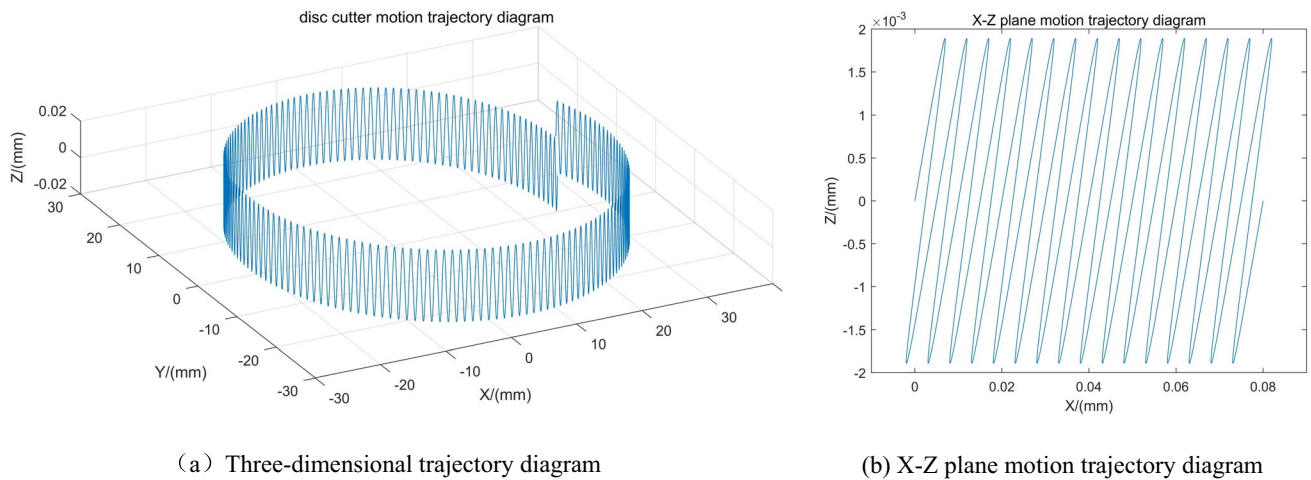
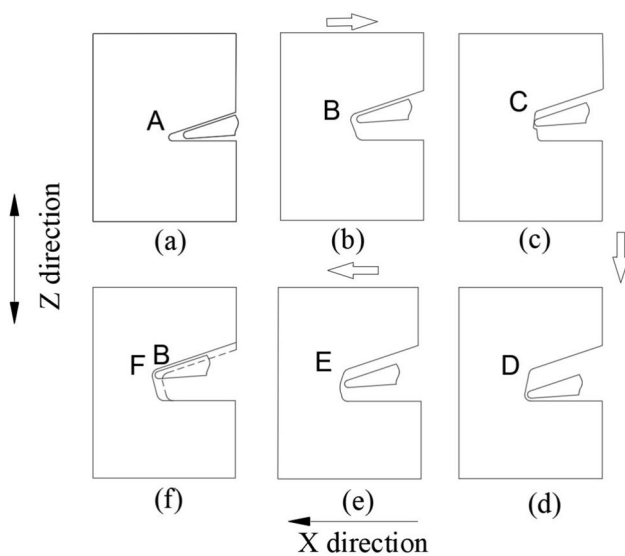


Fig. 2 Schematic diagram of disc cutter structure

Table 2 Disc cutter structure parameters (unit: mm)

Disc cutter	B_1	B_2	R	g	G	α
Value	12	20	25	1.4	7	14°

**Fig. 3** Trajectory of the disc cutter**Fig. 4** Schematic diagram of a disc cutter cutting honeycomb material in one cycle

pressure on the material, generating an extrusion force. The feed rate is very small at this point, so the amount of material removed during the tool cuts the material by upward extrusion is also very small. The cutting thickness gradually increases from point A to point B and reaches the maximum cutting thickness at point B. The ultrasonic vibration increases the speed of the tool, which can be considered as high-speed, low-feed cutting.

After arriving at point B, the direction of motion changes abruptly and the tool produces an impact force on the material, at which time the honeycomb wall will rebound downward due to its elastic deformation. The matrix material of Nomex honeycomb composite is aramid insulating paper, containing aramid precipitation fibers and aramid short-cut fibers, and there are many microcracks inside the material. The impact of the vibration direction on the material during the process of the disc cutter will produce upward tensile stress, causing the microcracks inside the material to expand under the action of tensile stress in the direction of the feed direction [19]. Then, the tool goes to point D; the front cutter face is separated from the chips; the tip of the tool squeezes the material downward, generating extrusion pressure. The cutting thickness gradually increases from point B to point D and reaches the maximum thickness when it reaches point D. At this time, the cutting thickness is the same as that at point B.

Subsequently, the tool proceeds from point D to point F, reaching the maximum cutting thickness at point F, repeating the cyclic cutting process. The contact process between the front cutter face of the disc cutter and the Nomex honeycomb material in a single cycle is shown in (f) of Fig. 3. During the upward motion of the cutter, when a point on the disc cutter's front face comes into contact with the highest point from the previous cycle (point B), the cutter's front face begins to compress the honeycomb material extensively, continuing until the cutter's tip reaches point F.

The cutting motion throughout the process is continuous cutting by the tool tip, with intermittent contact between

the tool face and the material. Compared to conventional cutting, ultrasonic vibration-assisted cutting reduces cutting forces and improves surface quality by reducing actual cutting thickness and increasing cutting speed.

3 Establishment of the cutting force expression for ultrasonic vibration-assisted cutting with a disc cutter

From the analysis in Section 2, it is known that the cutting form of ultrasonic vibration-assisted disc cutter cutting on honeycomb material is continuous cutting by the tool tip, with intermittent contact between the front cutter face and the material. The front cutter face of the disc cutter only contacts the material for a short time when it reaches the horizontal position at the previous cycle amplitude extreme and the next cycle extreme position, and each short contact generates a significant impact force on the cutter face. Due to the short contact time between the cutter face and the material, and the gap generated by high-speed vibration, a high-pressure air film is formed between the disc cutter tool and the honeycomb material, reducing the friction between the tool and the material. Therefore, frictional forces are not considered. Considering that the disc cutter has different motion forms in different directions, the cutting force of the disc cutter is decomposed into the Z directional cutting force in the vertical direction and the X directional cutting force in the feed direction. The theoretical formulas for the cutting force in these two directions are established to analyze the influence of various cutting parameters on the cutting force.

3.1 Establishment of the vertical cutting force model

The cutting force in the Z direction consists of three parts; one part is the extrusion force of the tool tip on the material, one part is the extrusion force generated by the contact between the cutter face and the material, and one part is the impact force generated by the impact action of ultrasonic vibration. Figure 5 shows the contact model between the disc cutter and the honeycomb wall at the extreme position. The shaded area represents the material removal area of the disc cutter during half a cycle, and it can be seen that only the upper part of the cutting edge is cutting the material. The point B is the contact position between the disc cutter and the honeycomb wall at the beginning of a cycle, and the F point is the contact position at the beginning of the next cycle. When the front cutter face of the disc cutter contacts the B point, the entire front face of the cutter is squeezed against the honeycomb wall until the tip reaches the F point

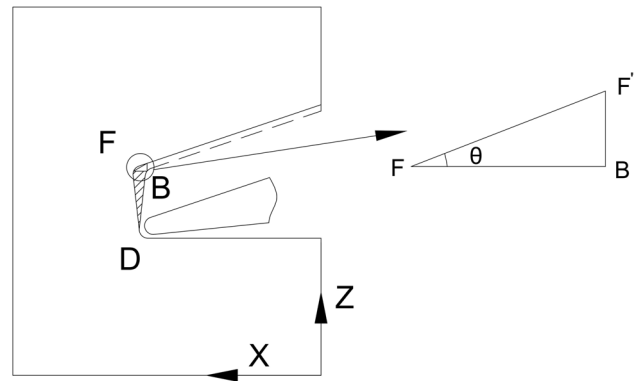


Fig. 5 Model diagram of removal of material cut by disc cutter

and stops moving upward, and the disc cutter starts to move downward.

where θ is the equivalent inclination angle of the disc cutter in the X–Z plane after the forward and lateral inclination of the disc cutter.

From Eq. (1), the area of the honeycomb material extruded by the tool tip in one cycle is:

$$S = \int_0^{x_i} Z_i dx \quad (2)$$

The duration of one cycle is:

$$t = T = \frac{2\pi}{\omega} = \frac{1}{f} \quad (3)$$

Considering the point on the disc cutter as a non-rotating point and neglecting its rotation, Eq. (1) can be brought in to get:

$$S = \frac{2AV_0 \cos \beta \cos \gamma}{\pi f} + A^2 \sin 2\beta \cos \gamma \quad (4)$$

The average width of the material extruded by the tip of the disc cutter during the entire process is

$$l = \frac{S}{A} = \frac{2V_0 \cos \beta \cos \gamma}{\pi f} + A \sin 2\beta \cos \gamma \quad (5)$$

From Eq. (5), the width of the material extruded by the tip of the tool increases with the increase of the feed rate and the vibration amplitude, but it is always smaller than the cutting width of the disc cutter, and when the disc cutter does not apply inclination angle, the second term in l does not exist and the value of l increases with the increase of V_0 .

$$l = \frac{2V_0}{\pi f} \quad (6)$$

When the inclination angle is applied, an additional term related to A is introduced. However, due to the increase in

cutting resistance and the potential for uneven force distribution and tool failure when the inclination angle is excessively large, a small inclination angle is typically applied, resulting in a negligible value of the second term. The value of l is mainly dominated by the first term.

The tip contact area S_{tip} can be derived from the geometric relationship in Fig. 6 and is twice as large as the shaded portion of the figure:

$$S_{tip} = \frac{\pi R^2}{180} \arccos\left(\frac{R}{R-l}\right) - \sqrt{R^2 - (R-l)^2}(R-l) \quad (7)$$

where l is the cutting width of the tip and R is radius of the disc cutter.

The geometrical relations of Fig. 5 show that:

$$F'B = FB \tan \theta \quad (8)$$

where FB is the distance traveled in the X direction by the disc cutter during the time it moves from point B to point F' in the Z direction.

$$FB = V_0 T \quad (9)$$

Then:

$$F'B = \frac{V_0}{f} \tan \theta \quad (10)$$

The distance traveled by the disc cutter in the Z direction of vibration at the same time is:

$$d = \int_{t'}^{t_F} 2A\pi f \cos(2\pi ft) \cos \beta \cos \gamma dt \quad (11)$$

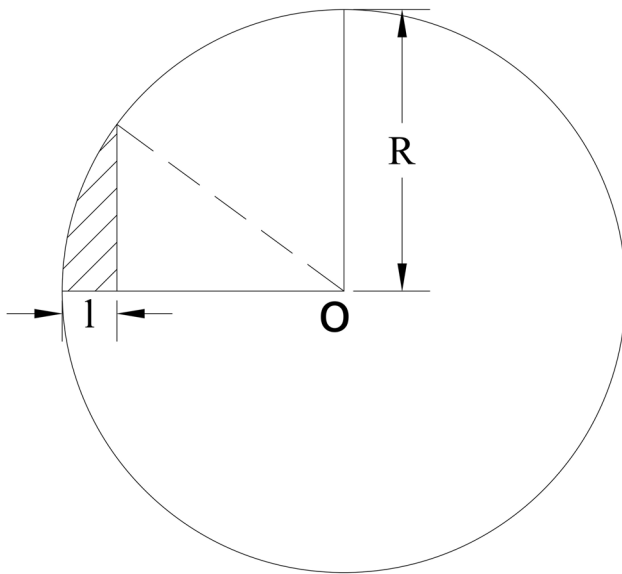


Fig. 6 Disc cutter geometry

where:

t' : The time when the disc cutter just contacts with the honeycomb wall.

t_F : The time when the disc cutter finishes contacting the honeycomb wall.

At the time when the front cutter face of the disc cutter leaves the honeycomb wall, the amount of rebound generated by the honeycomb wall is:

$$\varepsilon = \varepsilon' \cdot A \quad (12)$$

where ε' is the elastic strain of the Nomex honeycomb. It can be obtained:

$$\frac{V_0}{f} \tan \theta + \varepsilon' \cdot A = \int_{t'}^{t_F} 2A\pi f \cos(2\pi ft) \cos \beta \cos \gamma dt \quad (13)$$

Then:

$$t_F - t' = \frac{1}{\pi f} \left(\arcsin\left(\frac{V_0 \tan \theta}{A f \cos \beta \cos \gamma}\right) + \arcsin\left(\frac{\varepsilon'}{\cos \beta \cos \gamma}\right) \right) \quad (14)$$

$t_F - t'$ is the time when the front cutter face of the disc cutter extrudes the honeycomb wall in one cycle, and the tip of the cutter cuts the material for the rest of the time. Then, the contact time between the tip of the cutter and the material during one cycle is t' . It can be seen that as the feed rate V_0 increases, the value of $t_F - t'$ increases, and as the amplitude A increases, the value of $t_F - t'$ decreases.

The squeezing force generated when the material comes into contact with the tip of the disc cutter is:

$$F_{rZ1} = \frac{t'}{T} \sigma \cdot S_{tip} \quad (15)$$

where σ is the fracture stress of the material.

The average squeezing force generated when the material is in contact with the front cutter face is:

$$F_{rZ2} = \frac{t_F - t'}{T} \sigma \cdot S_f \quad (16)$$

where S_f is the cutting area of the front cutter face.

The squeezing force of the disc cutter on the material during the cutting process is:

$$F_{rZ} = F_{rZ1} + F_{rZ2} \quad (17)$$

The necessity for the cutting area of the tip of the disc cutter is always smaller than the cutting area between the front cutter face and the honeycomb wall. Therefore, the application of ultrasonic vibration leads to a reduced F_{rZ} compared to the without ultrasonic vibrated state. When the feed rate increases, according to Eq. (17), the time of $t_F - t'$ increases, resulting in higher F_{rZ2} and subsequently amplifying F_{rZ} . Similarly, an increase in cutting width causes F_{rZ2} to rise, consequently leading to a larger F_{rZ} . Consequently, the

cutting force in the Z direction of the disc cutter increases with the increase of feed rate and cutting width.

The velocity equation of the disc cutter can be obtained from Eq. (1):

$$V_Z = 2A\pi f \cos(2\pi ft) \cos \beta \cos \gamma \quad (18)$$

Acceleration equation in the Z direction of the disc cutter:

$$a_Z = -4A\pi^2 f^2 \sin(2\pi ft) \cos \beta \cos \gamma \quad (19)$$

The impact force generated when the disc cutter contacts the honeycomb material is:

$$F_{UZ\max} = ma = -4mA\pi^2 f^2 \cos(2\pi ft) \cos \beta \cos \gamma \quad (20)$$

where m is the mass of the disc cutter.

The average impact force of the ultrasonic vibration action is:

$$F_{UZ} = \frac{1}{T} \int_0^T F_{UZ\max} dt = 2A\pi f^2 \cos \beta \cos \gamma (\cos(2\pi ft) - 1) \quad (21)$$

The cutting force in the Z direction is the sum of the three components of the above cutting force:

$$F_Z = F_{rZ1} + F_{rZ2} + F_{UZ} \quad (22)$$

3.2 Theoretical formula of cutting force in feed direction

When establishing the theoretical formula for cutting force in the X direction, the Z direction cutting is not taken into account. At this time, the X direction cutting form under ultrasonic vibration can be analogized to the cutting form of a straight blade cutter under ultrasonic vibration. When ultrasonic vibration is used for assisted machining and a forward angle β is applied to the disc cutter, there is also an ultrasonic vibration component in the feed direction.

Therefore, the motion in the X direction is a combination of feed motion and ultrasonic vibration motion, and the cutting motion in the X direction is somewhat similar to that of a straight blade cutter. Since only the tip of the tool contacts the material in the X direction, the contact area between the tool and the material is very small, and ultrasonic vibration has a friction-reducing effect, so the frictional force between the tool tip and the material is not considered. The cutting force in the X direction is composed of two parts: the extrusion force of the material on the tool tip and the impact force of the ultrasonic vibration feed component. Figure 6 shows the motion trajectory of the X direction of the disc cutter.

It can be seen from Fig. 7 that the conditions for intermittent cutting of the disc cutter in the X direction are:

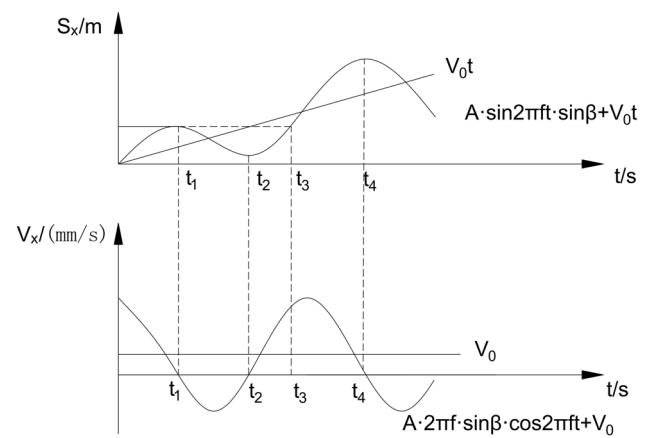


Fig. 7 Trajectory of the disc cutter in X direction

$$V_{X\min} < 0 \quad (23)$$

Assume:

$$V_0 < 2A\pi f \sin \beta \quad (24)$$

At time t_1 , the combined velocity V_1 of the disc cutter is zero, and the combined displacement is S_1 . At this time, the deepest point of the cutting into the material is reached. From t_1 to t_2 , the combined velocity is less than zero, and the cutter retreats in the X direction without cutting and moves away from the Nomex honeycomb composite material. At time t_2 , the combined velocity V_2 returns to zero, and the cutter tip is farthest away from the Nomex honeycomb composite material. From t_2 to t_3 , the tool tip continues to perform air cutting, but the combined velocity is greater than zero, and the tool tip moves towards the Nomex honeycomb composite material until at time t_3 , the combined displacement S_3 is equal to S_1 , and the tool tip recontacts the Nomex honeycomb composite material. From t_3 to t_4 , the tool tip starts cutting the Nomex honeycomb composite material again, and at time t_4 , the combined velocity V_4 of the cutter is zero, at which time the tool tip cuts into the deepest part of the Nomex honeycomb composite material.

Cutting time in one vibration cycle is:

$$t_s = t_4 - t_3 \quad (25)$$

At time t_4 , the combined velocity is 0:

$$V_0 + A \cdot 2\pi f \sin \beta \cdot \cos 2\pi f t_4 = 0 \quad (26)$$

The value of t_4 is:

$$t_4 = \frac{1}{2\pi f} \arccos \left(-\frac{V_0}{2A\pi f \sin \beta} \right) \quad (27)$$

The displacement of the tool from moment t_3 to moment t_4 is:

$$d_1 = V_0(t_4 - t_3) + A \sin \beta \sin \gamma (\sin 2\pi f t_4 - \sin 2\pi f t_3) \quad (28)$$

The feed in the cycle is:

$$d_2 = V_0 T = V_0 \cdot \frac{1}{f} \quad (29)$$

The displacement of the tool at the moment t_3 to t_4 is the feed of the tool movement in one cycle, so:

$$d_1 = d_2 \quad (30)$$

The relation between t_3 about t_4 and ultrasonic amplitude and feed rate can be obtained as follows:

$$V_0 \cdot \frac{1}{f} = V_0(t_4 - t_3) + A \sin \beta \sin \gamma (\sin 2\pi f t_4 - \sin 2\pi f t_3) \quad (31)$$

The velocity equation of the disc cutter in the X direction is:

$$V_X = 2A\pi f \cos 2\pi f t \cdot \sin \beta \quad (32)$$

The equation for the X direction acceleration of the disc cutter is:

$$a_X = -4A\pi^2 f^2 \sin(2\pi f t) \sin \beta \quad (33)$$

The impact force in the X direction is:

$$F_{UX \max} = m a_{X \max} = -4mA\pi^2 f^2 \sin \beta \quad (34)$$

The average impact force is:

$$\overline{F_{UX}} = \frac{1}{T} \int_{t_3}^{t_4} F_{UX \max} dt \quad (35)$$

The impact force generated when the disc cutter comes into contact with the upper and lower honeycomb walls has a tensile effect on the material, as shown in Fig. 8.

The tensile stress generated by impact vibration is

$$\sigma_t = \frac{F_{UX \max}}{S_f} = \frac{4mA^2\pi^2 f^2 \sin \beta}{S_f} \quad (36)$$

Considering the motion characteristics of the disc cutter, it can be known that the impact force of the disc cutter on the honeycomb wall during axial vibration produces a tensile stress wave. This tensile stress wave surpasses the fracture stress of the Nomex honeycomb. Consequently, the internal microcracks within the material expand under the influence of this tensile stress wave [4]. As a result, the material sustains internal damage, leading to a reduction in the force required for the material to fracture. The internal damage factor of the material is defined as D , and the larger

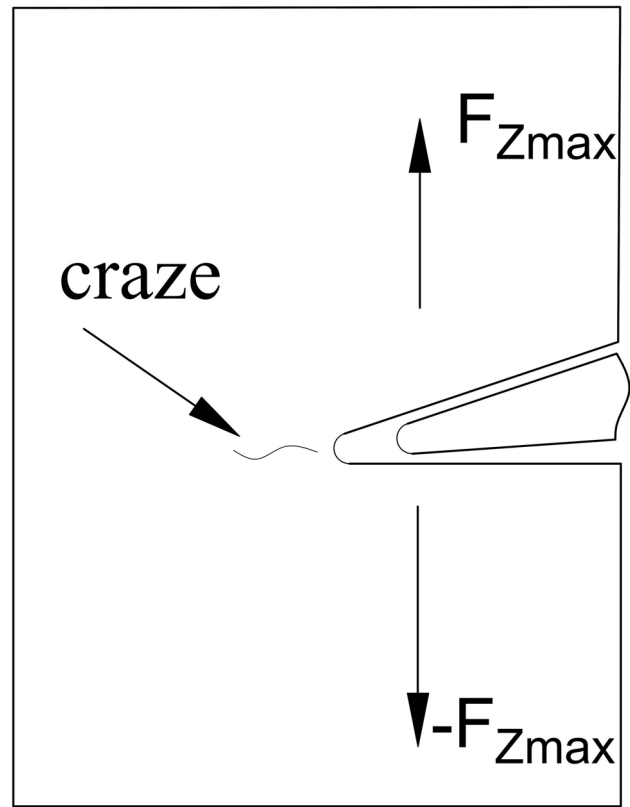


Fig. 8 The tensile stress diagram of the front material under impact loading

the tensile stress, the larger the damage factor D . As shown in Eq. (36), the value of D increases with the increase of the amplitude A and decreases with the increase of the cutting area of the front cutter face; that is, it decreases with the increase of the cutting width.

The actual fracture stress of the material is:

$$\sigma^* = \sigma \cdot (1 - D) \quad (37)$$

where σ is the nominal fracture stress.

The extrusion force of the material on the tool tip is:

$$F_{rX} = \sigma^* \cdot S_{tip} \quad (38)$$

The cutting force in X direction is:

$$F_X = F_{rX} + \overline{F_{UX}} \quad (39)$$

4 Experiment and data analysis

4.1 Experimental design

To verify the correctness of the cutting force theoretical model established in Section 3 and explore the influence of

various process parameters on cutting force, a single-factor experiment was designed. In each experiment, only one variable was changed while other process parameters were kept at the same level, so as to obtain the change rule of cutting force when a certain process parameter changes. The variables of this experiment are the ultrasonic amplitude, feed rate, and cutting width corresponding to the cutting force theoretical model. The unchanged machining parameters are ultrasonic vibration frequency f of 20,000 Hz, cutting depth of 3 mm, and spindle speed of 800 r/min. The forward angle and tilt angle of the circular disc cutter are both 1.5° . The single-factor experiment plan is shown in Table 3.

Table 3 Single-factor experiment plan for ultrasonic cutting with disc cutter

Group	Parameters		
	$A(\mu\text{m})$	$V_0(\text{mm}/\text{min})$	$e(\text{mm})$
A1	0	1200	10
A2	5	1200	10
A3	10	1200	10
A4	15	1200	10
A5	20	1200	10
A6	15	600	10
A7	15	1200	10
A8	15	1800	10
A9	15	2400	10
A10	15	1200	3
A11	15	1200	6
A12	15	1200	9
A13	15	1200	12

A total of 13 experiments were conducted, with 5 repetitions per group, and the average value was taken. A test bench for ultrasonic cutting of Nomex honeycomb materials with a disc cutter was set up as shown in Fig. 9. The ultrasonic vibration system consists of an ultrasonic digital power supply, an ultrasonic amplitude transformer, an ultrasonic transducer, and a disc cutter. The ultrasonic cutter handle is mounted on the spindle of the CNC machine tool, and the interface above the handle is connected to the ultrasonic digital power supply. The Nomex honeycomb sample is fixed on an aluminum plate by adhesive bonding, and the aluminum plate is fixed on a Kistler force measuring platform by bolts. The force measuring platform is fixed on the machine tool platform through a pressing plate.

During the cutting process of Nomex honeycomb, the stability of the ultrasonic machining system significantly affects the cutting force and the final surface quality. The instability factors within the ultrasonic machining system include the high-frequency vibration stability of the tool, resonance frequency, and tool amplitude. Therefore, it is necessary to conduct impedance testing and analysis on the vibration unit composed of the ultrasonic amplitude transformer, transducer, and cutting tool. In the experiment, an impedance analyzer was used for testing. The quality factor parameters of the ultrasonic vibration machining system were relatively high; the shape of the admittance circle was relatively intact, and the dynamic resistance was low. In the polar coordinate representation, a relatively complete “几” shape pattern was observed, indicating good impedance matching of the ultrasonic system. The results are shown in Fig. 10.

Fig. 9 Ultrasonic cutting experiment

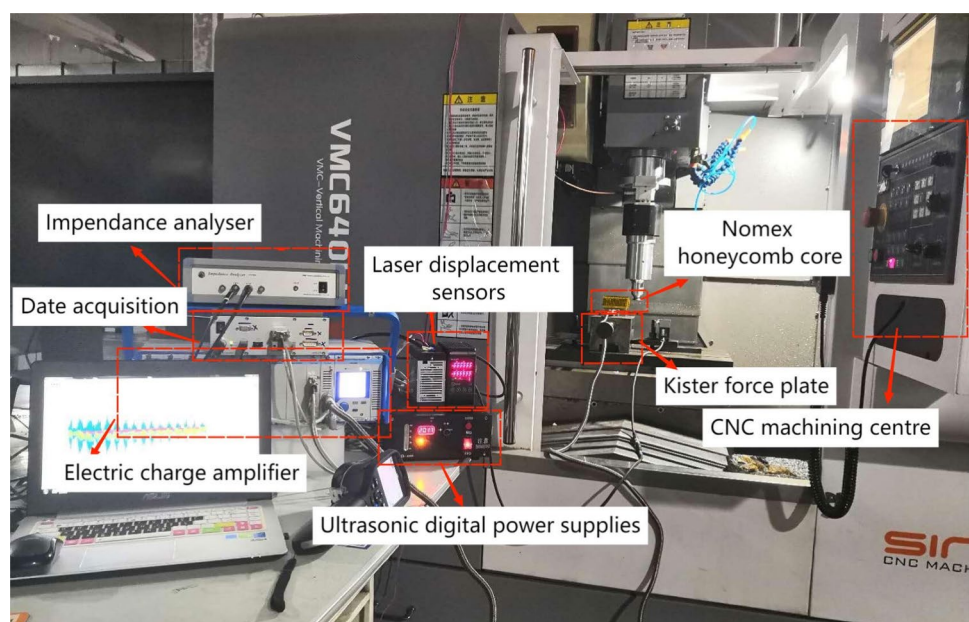


Fig. 10 Impedance analysis diagram

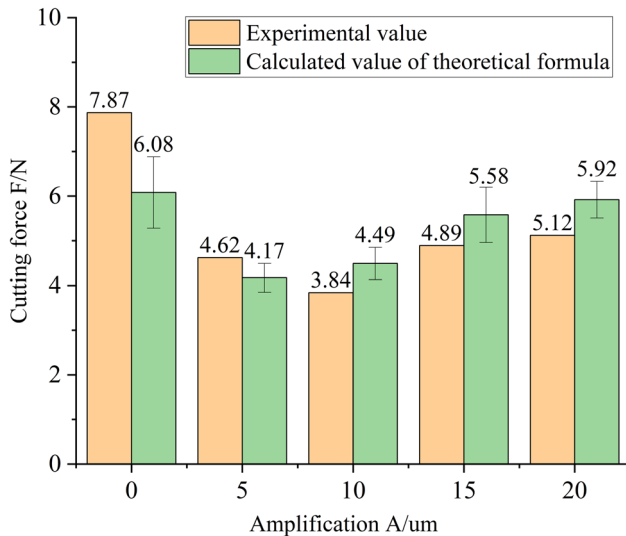
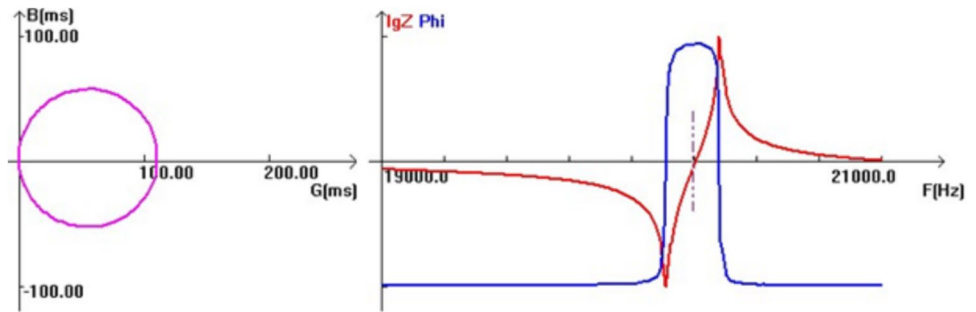


Fig. 11 The cutting force values in Z direction under different ultrasonic amplitudes

4.2 Data analysis of cutting force in the Z direction

An error analysis was performed by comparing the experimental cutting force data with the cutting force data calculated through theoretical formulas, with all parameters consistent with the experiment. The comparison between the experimental values of groups A1 to A5 and the corresponding theoretical formula values is shown in Fig. 11.

Without applying ultrasonic vibration, that is, when the amplitude is 0 μm , the cutting force is maximum. As the amplitude of ultrasonic vibration is increased, the cutting force first decreases and then increases. The experimental trend of cutting force change is consistent with the theoretical model, with an average error of 15.5%. As analyzed in Section 3, when ultrasonic vibration is applied, the cutting surface of the disc cutter no longer continuously compresses the honeycomb wall. The contact time between the cutting surface and the honeycomb wall within one vibration cycle is very short. According to Eq. (15), the contact time between the cutting surface and the honeycomb wall within one vibration cycle T can be calculated.

The cutting parameters in Eq. (15) are the same as those in the experiment, and the unit linear strain is the reciprocal of the elastic modulus of the honeycomb wall plate. The contact time between the front cutting surface and the honeycomb wall in one vibration cycle is about 1.52×10^{-4} T. The average impact force of ultrasonic vibration acting on the honeycomb wall is given by Eq. (21). Although the impact force generated by ultrasonic vibration is large, the average impact force is smaller than the compressive force of the front cutter face of the disc cutter continuously compressing the honeycomb wall due to the short contact time. Equations (14), (15), (16), and (21) in Section 3 can be used to analyze the effect of increasing ultrasonic amplitude on the contact time between the cutter face and the material, that is, the compressive force F_{rz1} exerted on the cutting edge and the compressive force F_{rz2} exerted on the cutting surface of the honeycomb wall, as well as the impact force F_{rz} generated by ultrasonic vibration. The increase of ultrasonic amplitude has a decreasing effect on the compressive force of the cutting edge, an increasing effect on the compressive force of the cutting surface, and a decreasing effect on the impact force. There exists an optimal ultrasonic amplitude for ultrasonic vibration cutting, which minimizes the sum of these two effects. This is also why too small or too large ultrasonic amplitudes will cause an increase in cutting force.

The analysis of experimental cutting force and the corresponding theoretical formula calculated cutting force for groups A6 to A9 are shown in Fig. 12.

From Fig. 12, it can be concluded that the cutting force increases with the increase of feed rate, and the average error between the theoretical formula calculation values and experimental values for different feed rates is 20.75%. The part explaining the effect of ultrasonic amplitude on cutting force illustrates the three components of the cutting process and cutting force in the Z direction. According to Eqs. (15) and (16), as the feed rate increases, the cutting area of the disc cutter tip and the front cutter face will both increase, resulting in an increase in the extrusion force of the cutter tip and the cutting face on the material. The impact force generated by ultrasonic vibration remains unchanged, resulting in an overall increase in cutting force.

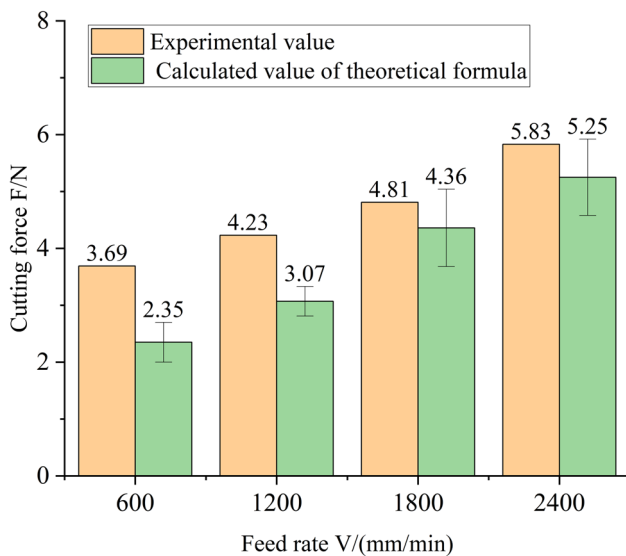


Fig. 12 The cutting force values in Z direction at different feed rates

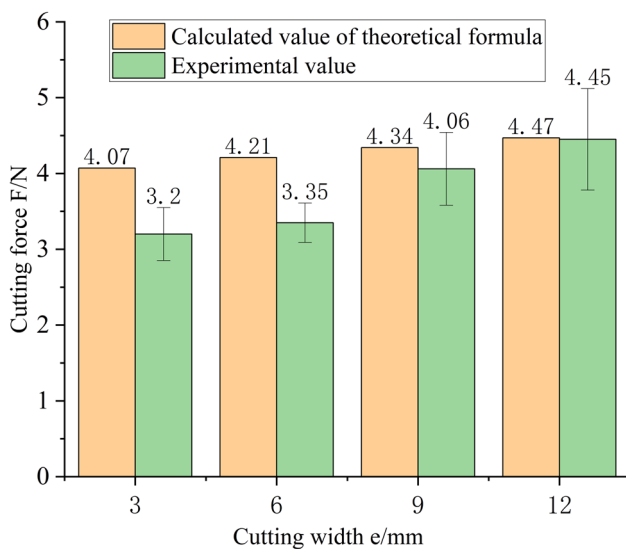


Fig. 13 The cutting force values in Z direction under different cutting widths

The analysis of the experimental cutting force of groups A10–13 and the corresponding cutting force calculated by theoretical formulas are shown in Fig. 13.

The average error between the theoretical formula calculation values and experimental values for different cutting widths in Fig. 12 is 12.17%. According to Eqs. (7) and (16), when the cutting width is changed, the cutting surface area of the front cutting surface of the disc cutter that extrudes the honeycomb material increases, resulting in an increase in extrusion force. The overall theoretical analysis in Section 3 shows that changing the cutting width has no effect on the extrusion force of the cutter tip and the vibration impact force. Therefore, when the cutting width increases, the overall cutting force increases (Table 4).

The model emphasizes the effects of ultrasonic amplitude and other cutting parameters on cutting forces, effectively explaining the reduction in cutting forces following the application of ultrasonic vibration and how these forces vary with changes in amplitude. During the modeling process, a time variable was introduced to divide the cutting process into two phases: the tool tip cutting phase and the tool face impact phase. The cutting force was analyzed as comprising three components: the tool tip cutting force, the tool face cutting force, and the vibration impact force, all represented through time functions that include various cutting parameters. However, this model is based on the premise that ultrasonic vibration alters the volume and form of material removal, neglecting surface stress and friction effects during the modeling process. Secondly, the absorptive honeycomb material wall panel is a multi-layer composite structure, with each layer consisting of different material components, including aramid paper, aramid fibers, and phenolic resin. Nevertheless, when establishing the theoretical cutting force model, it was treated as a homogeneous body, which could lead to certain inaccuracies in the calculated cutting forces. The resulting errors were found to be 15.5% in amplitude, 20.75% in feed rate, and 12.17% in cutting width, which are considered reasonable.

Table 4 Comparison of maximum difference between experiment and theoretical

	Cutting force model	Cutter	Vibration	Material	Maximum difference
Current study	Theoretical model	Disc cutter	Axial vibration	Nomex honeycomb	15.5% in amplitude, 20.75% in feed rate, and 12.17% in cutting width
Liu et al. [18]	Theoretical model	Milling cutter	Axial vibration	TiB2/7050Al	10% in Fx and Fy, 15% in Fz
Ahmad et al. [15]	Theoretical model	Disc cutter	Axial vibration	Nomex honeycomb	Qualitative comparison
Feng et al. [17]	Theoretical model	Milling cutter	Feed vibration	Aluminum alloy 2A12	13.6% in feed direction and 13.8% in cutting direction
Xiang [9]	Theoretical model	Straight edge cutter	Axial vibration	Nomex honeycomb	Qualitative comparison
Jinglu et al. [14]	Empirical model	Disc cutter	Axial vibration	Nomex honeycomb	10%

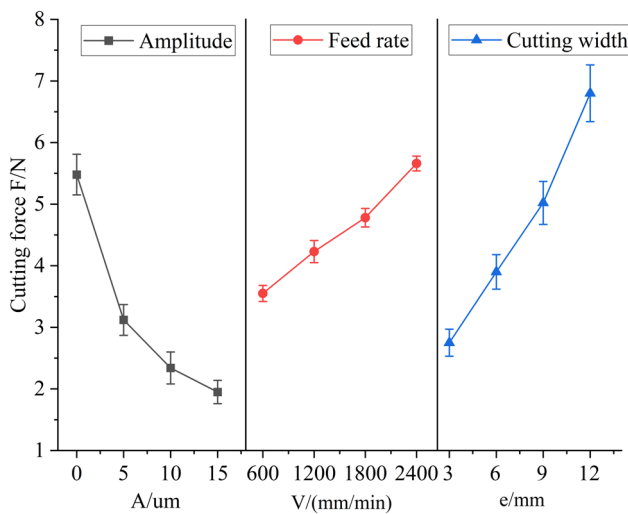


Fig. 14 The cutting force in X direction under different cutting parameters

4.3 Data analysis of cutting force in the X direction

From Fig. 14, It can be observed that the experimental results are consistent with the trends predicted by the theoretical model. The cutting force does indeed rise with an increase in cutting width, escalate with higher feed rates, and diminish with an augmentation in ultrasonic amplitude. As established in the theoretical analysis of the third section, augmenting the amplitude A contributes to an increase in the damage factor D and a decrease in the cutting force. When cutting width e increases, the damage factor D declines while the cutting force grows larger. Moreover, an increase in feed rate leads to a larger cutting layer area, resulting in heightened deformation resistance and, subsequently, an increase in the cutting force.

5 Conclusion

The cutting force theoretical formula established in this paper has directionality, which is different from the previous cutting force theoretical models obtained from a macroscopic level or empirical models based on experimental results. Starting from the tool path under ultrasonic vibration, the process of material removal at the microscopic level and the amount of removal within the vibration cycle were analyzed. The material removal amount was expressed by cutting process parameters and ultrasonic vibration parameters, and a cutting force theoretical model was established based on the material removal amount and ultrasonic impact. Different cutting force theoretical models were established for different vertical directions and feed directions, explaining the reasons for the reduction of

cutting force in these two directions after applying ultrasonic vibration from different perspectives. The ultrasonic disc cutting experiment of Nomex honeycomb material was carried out, and the experimental results were consistent with the trend of the cutting force theoretical formula results, verifying the accuracy of the mechanism analysis and cutting force theoretical formula. The main conclusions of this article are:

- (1) The removal model of Nomex honeycomb material by disc cutter cutting was altered after applying ultrasonic vibration. The ultrasonic cutting process with disc cutter involves continuous cutting with the tool tip and intermittent contact with the cutter face. This vibration mode has changed the way and pattern of material removal.
- (2) The impact force of ultrasonic vibration on the material's upper and lower surfaces induces tensile stress in the uncut portion ahead. This stress expansion within the uncut area leads to internal microcrack propagation and subsequent material damage. Consequently, the material's fracture stress is diminished, resulting in a reduction in cutting force.
- (3) Concerning the Z direction cutting force, as the feed rate rises, the contact area between the tool tip and the material expands. Concurrently, the duration of contact between the disc front cutter face and the honeycomb wall increases, leading to an escalation in cutting force. Similarly, when the cutting width increases, the contact area between the disc front cutter face and the honeycomb wall enlarges. This results in a greater extrusion force from the front cutter face onto the honeycomb wall, subsequently causing an increase in cutting force.
- (4) In terms of the X direction cutting force, an increase in amplitude prompts greater internal material damage, leading to a reduction in cutting force. As for the effect of cutting width and feed rate, an increase in both parameters results in a larger cutting layer area, subsequently raising deformation resistance and causing an increase in cutting force.

Author contribution Huawei Ji, Jiace Wu, and Xiaoping Hu planned the research. Jiace Wu and Weijun Hong did the experiments. Jiace Wu wrote the paper and analyzed the data. Huawei Ji and Xiaoping Hu revised the paper.

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Data availability This article contains all the data gathered or analyzed during this study.

Declarations

Ethical approval None of the studies mentioned in this article contains any human participation. Also, no animals were harmed during these experiments.

Consent to participate The authors consent to participate.

Consent for publication The authors provide their consent to publish this article.

Competing interests The authors declare no competing interests.

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